Final Report

Image Analysis of Park Infrastructure

Date:	2026-06-25
Prepared for:	BC Parks
Prepared by:	Sarah Gauthier
	Nour Shawky
	Ssemakula Peter Wasswa
	Mehmet Imga

Table of contents

1 Executive summary

2 Introduction

2.1 Background and problem statement

2.2 Refined objectives

3 Data

3.1 Data sources and collection

3.2 Size and characteristics

3.3 Data quality, limitations, and preprocessing

4 Data science methods

4.1 Approaches considered

4.2 VLM Approach

4.3 Embedding-based classification

The second modelling path treated each image as input to a pre-trained vision encoder and used the resulting embedding as a fixed feature representation. The motivation was practical: our labelled dataset is useful, but not large enough to train a high-capacity image model from scratch. Instead, we used foundation models that had already learned general visual representations from large-scale pretraining, froze their weights, and trained lightweight classifiers on top of their embeddings. This gave us a supervised image-based approach while keeping the number of trainable parameters small and the experiments reproducible.

The embedding pipeline was the same for each vision backbone. First, images were loaded from the cleaned local image directory and passed through a frozen image encoder. Each photo was converted into a dense numeric vector. Because many BC Parks assets have multiple photos, we then averaged the image-level vectors for the same `asset_id` to create one asset-level embedding. This asset-level table was joined to the attribute-specific training labels, and one classifier was trained per target attribute. This design avoided duplicating labels across multiple photos while still allowing all available visual evidence for an asset to contribute to the final representation.

We evaluated three embedding families:

- **DINOv3**: a self-supervised vision transformer trained to produce general visual features without relying on image-text labels (Simeoni et al. 2025). We used DINOv3 as the main representation learner because self-supervised visual features are often strong for object shape, structure, texture, and material cues, all of which are central to infrastructure attributes.
- **SigLIP**: a vision-language model trained with a sigmoid image-text pretraining loss (Zhai et al. 2023). We used only the vision encoder, not the text encoder. The reason for including SigLIP was that language-aligned pretraining can sometimes encode higher-level semantic scene information that is useful for attributes such as structure position or decking material.
- **OpenCLIP / CLIP-style embeddings**: a contrastive image-text baseline based on the CLIP family of models (Radford et al. 2021). This provided an additional comparison point for language-aligned embeddings.

For the classifier layer, we tested logistic regression, linear support vector machines, random forests, and histogram-based gradient boosting using scikit-learn (Pedregosa et al. 2011). Logistic regression was especially important because a strong linear classifier indicates that the embedding space already separates the attribute classes reasonably well. We also tested a tuned logistic-regression variant by searching over regularization strength and class weighting inside the training folds. In practice, the tuned variant did not consistently improve over the simpler default logistic-regression setup, so the untuned logistic model remained the main embedding classifier.

All embedding classifiers were evaluated with grouped cross-validation by `asset_id`. This was a key design choice. If photos from the same asset appeared in both training and validation folds, the model could appear to generalize while actually recognizing near-duplicate visual evidence from the same physical asset. Grouping by asset ensured that every validation prediction was made for an asset that was not present in that fold's training set. Where possible, we used stratified grouped folds to preserve class proportions; otherwise, we used grouped folds without stratification when rare classes made stratification infeasible. For binned physical attributes whose labels can mean different things across asset types, the implementation also supports fitting separate models by asset type and pooling the out-of-fold predictions for summary metrics.

We used **weighted F1** as the primary model-selection metric because the attribute labels are highly imbalanced. A metric that only rewards the dominant class would overstate model quality for attributes where one material or one structural category appears much more often than the others. Weighted F1 balances precision and recall while weighting each class by its support, making it a better summary of performance on the distribution of labels BC Parks is most likely to encounter. Macro F1 was retained only as a diagnostic for minority-class behaviour because it gives each class equal weight regardless of frequency.

The overall pattern from the embedding experiments was that frozen embeddings substantially outperformed the majority-class baseline, and the best performing configuration was generally **DINOv3 embeddings with logistic regression**. SigLIP was competitive and was strongest for a small number of more context-dependent attributes, while tree-based classifiers had less consistent weighted-F1 performance across attributes. The average cross-validation scores from the main embedding runs are summarized below, ordered by mean weighted F1.

Table 1: Average grouped cross-validation performance for the main embedding-based classifiers. Weighted F1 was used as the primary comparison metric.

Model configuration	Attributes evaluated	Mean weighted F1	Mean macro F1
DINOv3 + logistic regression	12	0.749	0.479
SigLIP + logistic regression	12	0.742	0.462
OpenCLIP + logistic regression	12	0.727	0.443
Baseline majority class	12	0.540	0.201

These results suggest that the strongest image-only signal came from DINOv3's self-supervised visual representation: it produced the highest mean weighted F1 among the embedding approaches and a large gain over the majority-class baseline. At the same time, the attribute-level results showed that no single embedding family was uniformly best. SigLIP performed particularly well for `attr_structure_position` and `attr_decking_material`, suggesting that language-aligned embeddings may capture some contextual cues that complement DINOv3. For the final modelling discussion, we therefore treat DINOv3 + logistic regression as the primary embedding-based classifier and SigLIP/OpenCLIP as useful comparative baselines.

5 Data product and results

5.1 The Data Product

5.2 Results

6 Conclusions and recommendations

References

Pedregosa, Fabian, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, et al. 2011. "Scikit-Learn: Machine Learning in Python." *Journal of Machine Learning Research* 12: 2825–30. https://jmlr.org/papers/v12/pedregosa11a.html.

Radford, Alec, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, et al. 2021. "Learning Transferable Visual Models from Natural Language Supervision." In *Proceedings of the 38th International Conference on Machine Learning*, 8748–63. https://arxiv.org/abs/2103.00020.

Simeoni, Oriane, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, et al. 2025. "DINOv3." *arXiv Preprint arXiv:2508.10104*. https://arxiv.org/abs/2508.10104.

Zhai, Xiaohua, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. 2023. "Sigmoid Loss for Language Image Pre-Training." *arXiv Preprint arXiv:2303.15343*. https://arxiv.org/abs/2303.15343.